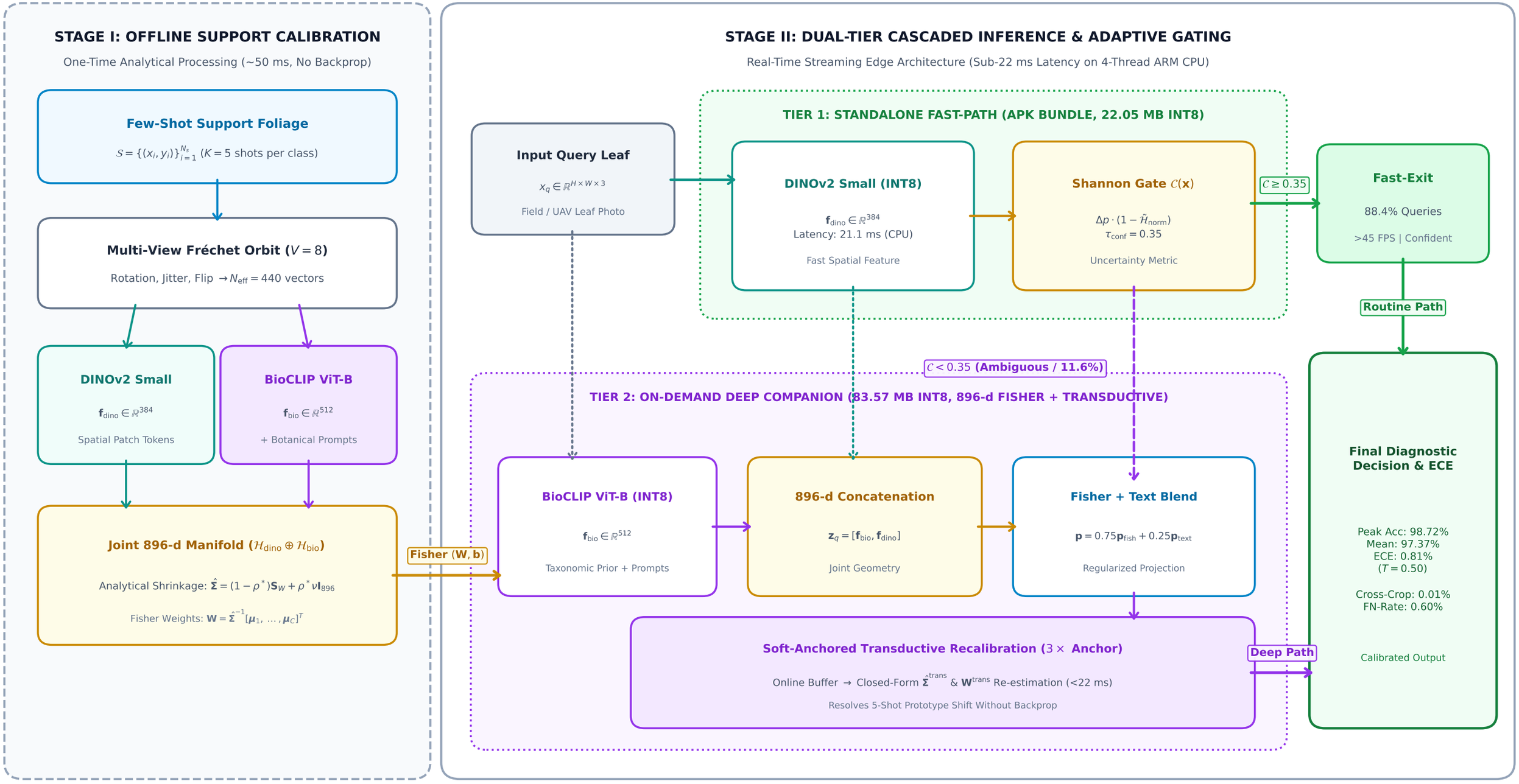

STAGE I: OFFLINE SUPPORT CALIBRATION
One-Time Analytical Processing (~50 ms, No Backprop)
Few-Shot Support Foliage
$\mathcal{S} = \{(x_i, y_i)\}_{i=1}^{N_s}$ ($K = 5$ shots per class)
Multi-View Fréchet Orbit ($V = 8$)
Rotation, Jitter, Flip $\rightarrow N_{\text{eff}} = 440$ vectors
DINOv2 Small
$\mathbf{f}_{\text{dino}} \in \mathbb{R}^{384}$
Spatial Patch Tokens
BioCLIP ViT-B
$\mathbf{f}_{\text{bio}} \in \mathbb{R}^{512}$
+ Botanical Prompts
Joint 896-d Manifold ($\mathcal{H}_{\text{dino}} \oplus \mathcal{H}_{\text{bio}}$)
Analytical Shrinkage: $\hat{\mathbf{\Sigma}} = (1-\rho^*)\mathbf{S}_W + \rho^* \nu \mathbf{I}_{896}$
Fisher Weights: $\mathbf{W} = \hat{\mathbf{\Sigma}}^{-1}[\boldsymbol{\mu}_1, \ldots, \boldsymbol{\mu}_C]^T$
Fisher ($\mathbf{W}, \mathbf{b}$)
STAGE II: DUAL-TIER CASCADED INFERENCE & ADAPTIVE GATING
Real-Time Streaming Edge Architecture (Sub-22 ms Latency on 4-Thread ARM CPU)
TIER 1: STANDALONE FAST-PATH (APK BUNDLE, 22.05 MB INT8)
Input Query Leaf
$x_q \in \mathbb{R}^{H \times W \times 3}$
Field / UAV Leaf Photo
DINOv2 Small (INT8)
$\mathbf{f}_{\text{dino}} \in \mathbb{R}^{384}$
Latency: 21.1 ms (CPU)
Fast Spatial Feature
Shannon Gate $\mathcal{C}(\mathbf{x})$
$\Delta p \cdot (1 - \tilde{\mathcal{H}}_{\text{norm}})$
$\tau_{\text{conf}} = 0.35$
Uncertainty Metric
$\mathcal{C} \geq 0.35$
Fast-Exit
88.4% Queries
>45 FPS | Confident
Routine Path
$\mathcal{C} < 0.35$ (Ambiguous / 11.6%)
TIER 2: ON-DEMAND DEEP COMPANION (83.57 MB INT8, 896-d FISHER + TRANSDUCTIVE)
BioCLIP ViT-B (INT8)
$\mathbf{f}_{\text{bio}} \in \mathbb{R}^{512}$
Taxonomic Prior + Prompts
896-d Concatenation
$\mathbf{z}_q = [\mathbf{f}_{\text{bio}}, \mathbf{f}_{\text{dino}}]$
Joint Geometry
Fisher + Text Blend
$\mathbf{p} = 0.75\mathbf{p}_{\text{fish}} + 0.25\mathbf{p}_{\text{text}}$
Regularized Projection
Soft-Anchored Transductive Recalibration (3× Anchor)
Online Buffer → Closed-Form $\hat{\mathbf{\Sigma}}^{\text{trans}}$ & $\mathbf{W}^{\text{trans}}$ Re-estimation (<22 ms)
Resolves 5-Shot Prototype Shift Without Backprop
Deep Path
Final Diagnostic Decision & ECE
Peak Acc: 98.72%
Mean: 97.37%
ECE: 0.81%
($T = 0.50$)
Cross-Crop: 0.01%
FN-Rate: 0.60%
Calibrated Output